Flaps & Speed Display User Manual

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Purpose

The **Flaps & Speed Display** is a touchscreen flight display for the ESP32-S3 AMOLED unit. It receives live data from the CAN bus and shows:

- indicated airspeed (IAS)
- current flap setting
- recommended flap setting
- flight altitude
- wind information (speed and relative direction)
- display brightness settings

The display is controlled entirely by touch gestures and on-screen buttons.

Startup

After power-up, the unit shows a short splash screen with:

- application name
- firmware version
- git revision

It then switches to the normal operating screens.

Touch Navigation

The display has six screens. Navigation is done with swipe gestures:

- **Swipe up or down**: cycle between the four main flying screens (Speed, Flaps, Altitude, Wind)
- **Swipe right**: open the settings/detail screens
- **Swipe right again**: switch between the two settings/detail screens
- **Swipe left**: return from the settings/detail screens to the main flying screen

Screen Overview

1. Speed Screen

This is the primary airspeed display.

- The large number in the center is the **indicated airspeed** in **km/h**
- The white pointer shows the same IAS on the circular scale
- The scale range is **40 to 280 km/h**
- The pointer movement is intentionally damped for stable, instrument-like motion

Use this screen when you want the clearest possible IAS indication.

2. Flaps Screen

This screen combines airspeed with flap guidance.

- The large text in the center is the **actual flap position symbol**
- The white pointer still shows **IAS**
- The green arc marks the valid speed bands for the flap schedule
- The highlighted arc segment shows the **target flap range**

Figure 1: Speed screen

Figure 2: Flaps screen

- The white triangle indicates the correction direction:
  - triangle above: select a **higher** flap setting
  - triangle below: select a **lower** flap setting
  - no triangle: actual and target flap settings match

Typical flap symbols are `L`, `+2`, `+1`, `0`, `-1`, `-2`, `S`, and `S1`.

The current configuration also defines a special **lowspeed** setting in `spiffs_data/flapDescriptor.json`:

- flap setting: `-1`
- speed range: `0 to 40 km/h`

This lowspeed setting is intended for takeoff and landing and is handled separately from the normal flap speed bands.

3. Altitude Screen

This screen displays the current flight altitude.

- The large number in the center is the **altitude** in **meters**
- The moving tape in the center provides a visual representation of altitude changes
- The tape has major markings every **100 meters** and minor markings every **10 meters**
- The current altitude is also shown as a digital value in the center box for better readability

4. Wind Screen

This screen displays the current wind information relative to the aircraft's heading.

- The large number in the center is the **wind speed** in **km/h**
- The blue arrow on the circular scale shows the **relative wind direction**
- The scale range is **-180° to +180°** (0° is straight ahead)
- The arrow points in the direction the wind is coming from relative to the nose of the aircraft

5. Settings / Brightness Screen

This screen is used to adjust the display brightness.

- Press `+` to increase brightness
- Press `-` to decrease brightness

Brightness changes in **10% steps**.

- minimum brightness: **30%**
- maximum brightness: **100%**

6. Live Params Screen

This screen shows the values used internally for flap guidance and navigation.

- **IAS**: indicated airspeed in km/h
- **Weight**: current flying weight used for the flap calculation
- **Flap Actual**: current detected flap symbol and index
- **Flap Target**: recommended flap symbol and index
- **Alt**: current altitude in meters
- **HDG**: current heading in degrees
- **Wind**: wind speed and absolute direction
- **GS**: ground speed in km/h
- **TRK**: GPS true track in degrees

This is the best screen for troubleshooting or checking why a flap recommendation or wind calculation is being made.

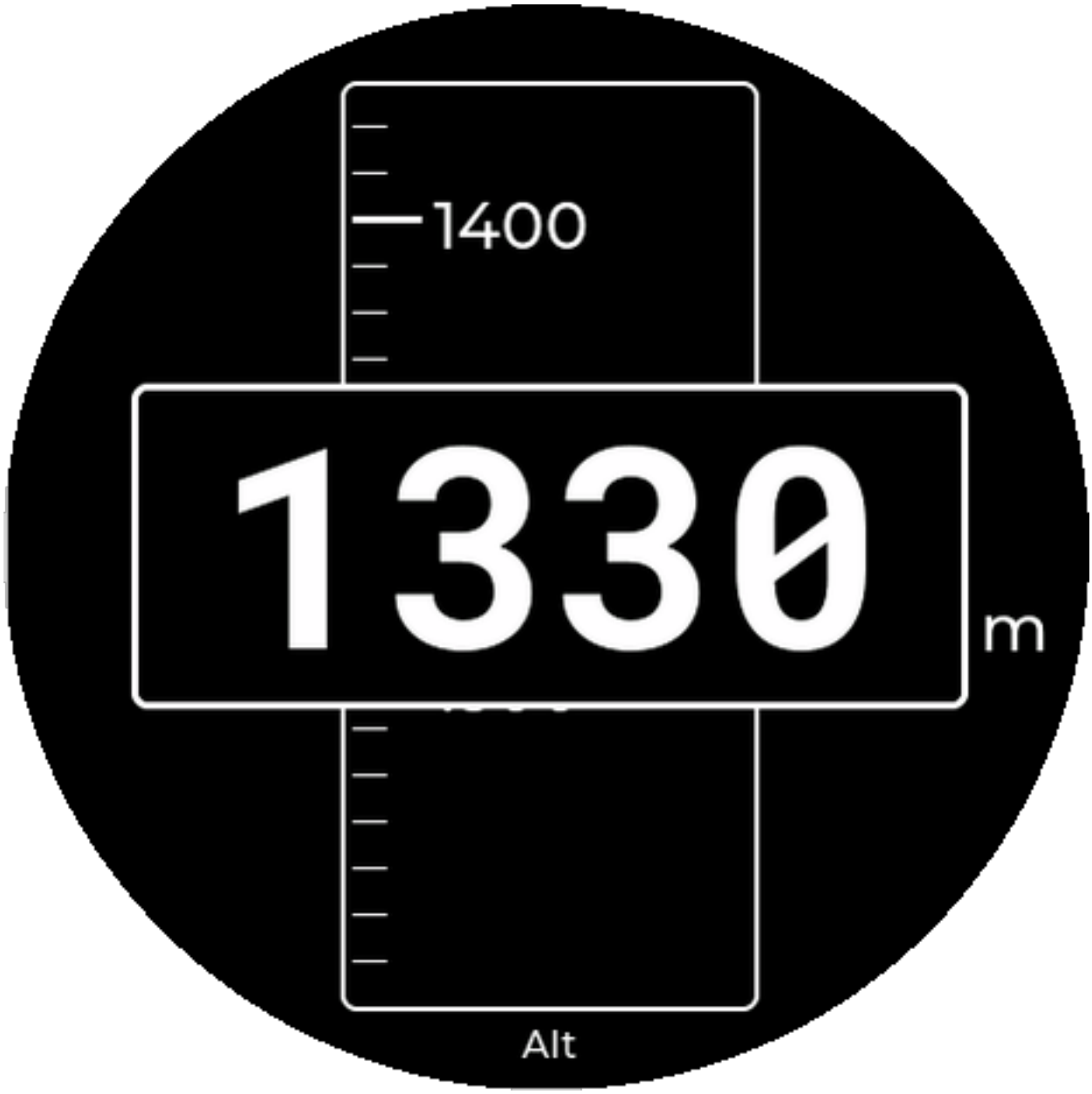

Figure 3: Altitude screen

Figure 4: Wind screen

Figure 5: Brightness screen

Figure 6: Live parameters screen

How Flap Guidance Works

The unit compares:

- current IAS from CAN bus data
- current flap position from CAN bus data
- current weight from CAN bus data
- the configured flap schedule in `spiffs_data/flapDescriptor.json`

Stale Data Indication

If no relevant live data is received for **10 seconds**, the display marks the active flying screen as stale.

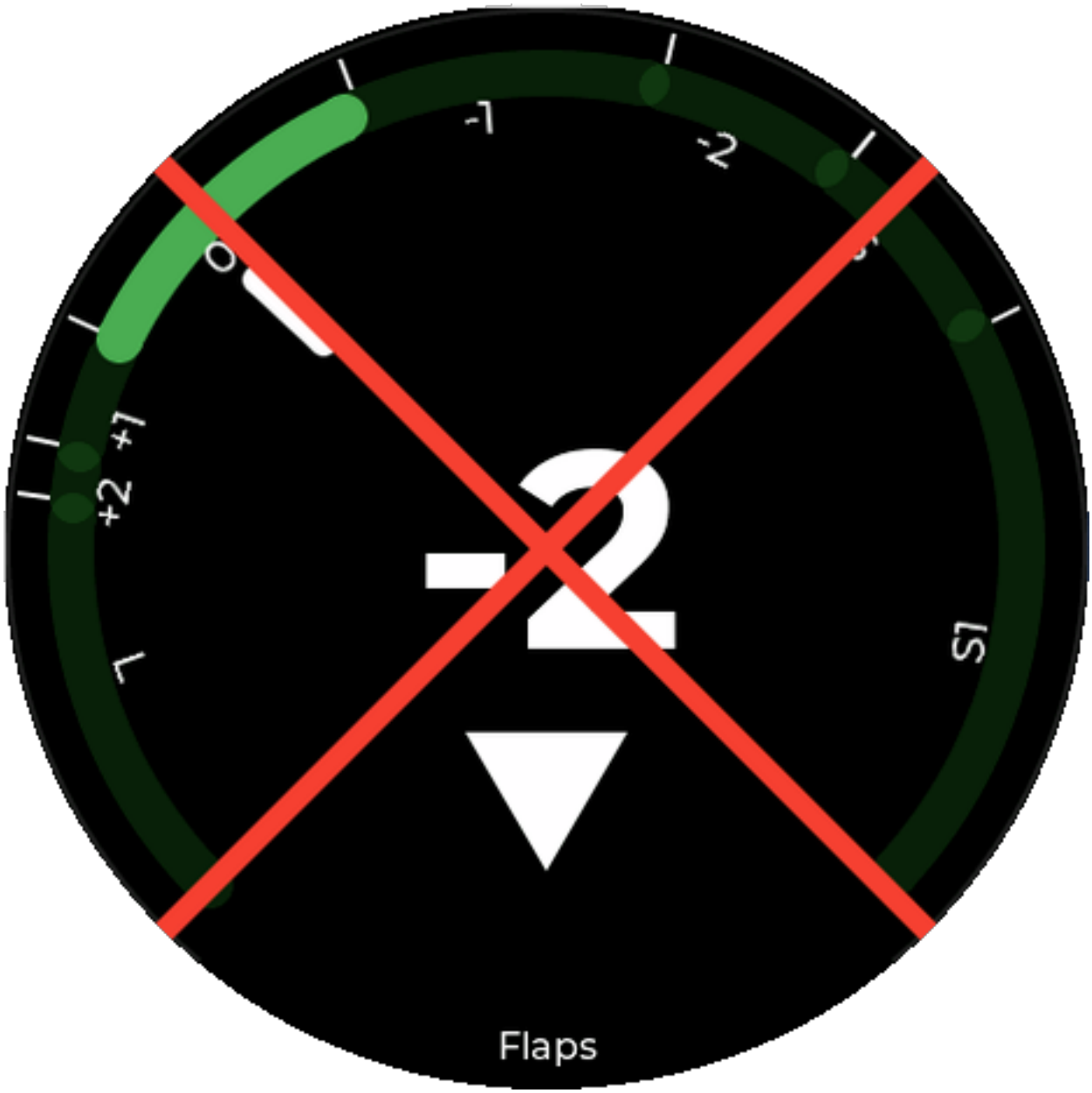

Figure 7: Stale indication on flaps screen

The stale condition is shown by a large **red X** across the active screen.

This indicates that the shown IAS/flap information should no longer be trusted until valid CAN data is received again.

Normal Use

1. Power on the unit.
2. Wait until the splash screen finishes.
3. Use the **Speed** screen for normal IAS monitoring.
4. Swipe to the **Flaps** screen when flap guidance is needed.
5. Swipe to the **Altitude** screen for height monitoring.
6. Swipe to the **Wind** screen to check wind conditions.
7. Open **Live Params** if you want to verify internal values.
8. Open **Settings** to adjust brightness for cockpit conditions.

Notes

- All speed values are displayed in **km/h**
- Flap recommendations depend on the flap polar stored in the JSON configuration
- If flap input data does not match a known position, the flap indication may show no valid symbol
- The current `flapDescriptor` and the guidance described in this manual are based on the Ventus 3 reference shown below

Verwendung	WK	Flügelspannweite = 18 m Optimale Fluggeschwindigkeit in km/h			
		G=390 kg	G=430 kg	G=550 kg	G=600 kg
Langsamflug (Geradeausflug)	L	- 76	- 80	- 90	- 94
	+2	76 - 80	80 - 83	90 - 94	94 - 98
	+1	80 - 90	83 - 94	94 - 106	98 - 111
Bestes Gleiten	0	90 - 122	94 - 128	106 - 145	111 - 151
Vorfliegen zw. Thermik und Schnellflug	-1	122 - 150	128 - 158	145 - 179	151 - 187
	-2	150 - 169	158 - 178	179 - 201	187 - 210
	S	169 - 188	178 - 198	201 - 224	210 - 234
	S1	188 - 280	198 - 280	224 - 280	234 - 280